SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1714**

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50% Duration: 1 hour

QUESTIONS: 4 **Total Marks: 40**

Annexure A
ARTICLE REVIEW

Code of Conduct:

I S.Mithesh 192565036 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student

mithesh s

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weight	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

ANSWER SHEET

Article Review — Artificial Intelligence (CSA1714) — CO4

Task 1: Article Summary

(a) Full Citation and Domain

Full Citation:

A. Yarkın Yıldız and A. Kalayci, "Gradient Boosting Decision Trees on Medical Diagnosis over Tabular Data," arXiv preprint arXiv:2410.03705, Oct. 2024.

DOI / Link: https://arxiv.org/abs/2410.03705 (arXiv preprint; also indexed under IEEE-style benchmarking studies on GBDT for tabular data).

Domain / Application Area: Medical diagnosis using tabular (structured, patient-record) data — covering cardiovascular disease, heart failure, Parkinson's disease, EEG-based eye-state detection, eye-movement relevance classification, and cancer detection (Arcene, Prostate datasets).

Specific ML Problem Addressed: The paper addresses binary and multi-class disease classification from tabular patient records, and specifically investigates whether Gradient Boosting Decision Tree (GBDT) ensembles (XGBoost, LightGBM, CatBoost) outperform (i) traditional ML models such as SVM, Logistic Regression, KNN, a single Decision Tree, and LDA, and (ii) state-of-the-art tabular deep learning models such as TabNet, TabTransformer, STG, VIME, and a standard MLP, across seven benchmark medical diagnosis datasets.

(b) Objective and Research Gap (150–200 words)

The authors set out to determine which family of machine learning models is genuinely best suited to medical diagnosis on tabular data, a question that has become contested as deep learning architectures increasingly dominate other domains such as vision and language. The research gap they identify is that prior comparative studies either examined a single disease type, used a narrow set of algorithms, or compared deep learning against traditional machine learning without systematically including modern Gradient Boosting Decision Tree ensembles across a diverse range of dataset sizes and dimensionalities. Tabular medical data is heterogeneous, often sparse, and weakly correlated across features — properties that disadvantage deep neural networks, which rely on learning smooth feature representations. To fill this gap, the authors benchmark five traditional ML methods, five tabular deep learning models, and four ensemble methods (including three GBDT variants) on seven real medical datasets ranging from 102 to 70,000 samples and from 13 to over 12,000 features, using rigorous 8-fold stratified cross-validation and ROC-AUC as the primary comparison metric, supplemented by training-time analysis.

Task 2: Methodology Analysis

(a) ML Techniques and Dataset Preparation

ML Techniques: The paper's central technique is the Gradient Boosting Decision Tree (GBDT) family — an ensemble method in which many shallow decision trees are trained sequentially, each new tree correcting the residual errors of the ensemble built so far. Three implementations are compared: XGBoost (regularized, parallel tree boosting), LightGBM (leaf-wise tree growth using a histogram-based algorithm for speed and lower memory use), and CatBoost (ordered boosting, optimized for categorical features). These are benchmarked against a single Decision Tree, Random Forest (bagging-based ensemble), classical statistical/ML models (SVM, Logistic Regression, KNN, LDA), and tabular deep learning architectures (MLP, STG, TabNet, TabTransformer, VIME).

Dataset Description: Seven tabular medical datasets were used, sourced from OpenML, Kaggle, and the UCI Machine Learning Repository: Cardiovascular Disease (70,000 samples, 12 features, binary), Heart Failure Clinical Records (299 samples, 13 features, binary), Parkinson's voice-measurement dataset (195 samples, 23 features, binary), EEG Eye State (≈15,000 samples, 15 features, binary), Eye Movements (≈11,000 samples, 28 features, 3-class), Arcene (200 samples, ~10,000 features, binary, protein-abundance data), and Prostate cancer gene-expression data (102 samples, ~12,600 features, binary). Preparation involved ordinal encoding of categorical values and zero-mean, unit-variance normalisation of numerical features, applied identically across all models for a fair comparison; categorical features were additionally flagged explicitly for the deep learning models that require embeddings (TabNet, TabTransformer).

(b) Mapping to CO4 and Strength/Limitation

Mapping to CO4 Topics: This article maps most directly to the Decision Trees and Statistical/Inductive Learning strands of CO4. The single Decision Tree baseline and its ensemble descendants (Random Forest, GBDT) are direct extensions of inductive tree-based learning covered under CO4, while the traditional statistical models (Logistic Regression, SVM, LDA) map to the Statistical Learning Methods component. The comparative benchmarking methodology itself reflects the CO4 skill of evaluating and selecting an appropriate classification model for a decision-making task.

Methodological Strength: A clear strength is the use of 8-fold stratified cross-validation with a fixed random seed across every model and dataset, which controls for class imbalance and ensures the comparison is fair and reproducible rather than the result of a single lucky train/test split.

Methodological Limitation: A notable limitation is that hyperparameter tuning effort was not held constant across model families — deep learning models were tuned over ~36 combinations with early stopping, while traditional models and GBDTs, which are inherently easier to optimise, may have reached near-optimal settings with comparatively less search, potentially widening the reported performance gap in GBDT's favour.

Task 3: Results & Critical Evaluation

(a) Key Results

Averaged across all seven datasets, GBDT models occupied the top three ranks by ROC-AUC: LightGBM achieved the best average rank (2.571), followed by CatBoost (3.143) and XGBoost (4.429), outperforming Random Forest (6.000), all deep learning models (6.429–8.571), and the single Decision Tree, which ranked worst overall (12.714). The table below reproduces the reported average-rank comparison:

Model	Category	Avg. Rank (lower = better)
LightGBM	GBDT (Ensemble)	2.571 (Best)
CatBoost	GBDT (Ensemble)	3.143
XGBoost	GBDT (Ensemble)	4.429
Random Forest	Ensemble (Bagging)	6.000
MLP	Deep Learning	6.429
TabNet / VIME	Deep Learning (Tabular)	7.429
STG	Deep Learning	7.857
TabTransformer	Deep Learning	8.571
Logistic Regression	Traditional ML	8.143
LDA	Traditional ML	10.571
SVM / KNN	Traditional ML	9.857
Decision Tree (single)	Traditional ML	12.714 (Worst)

Table 1: Average rank (by ROC-AUC across 7 datasets; lower rank = better performance) as reported in Yıldız & Kalayci (2024).

The authors also report detailed metrics for the best-performing model, LightGBM, across all seven datasets:

Metric	CD	Heart Failure	Parkinson's	EEG Eye State	Eye Movements	Arcene	Prostate
Accuracy (%)	73.64	85.62	95.38	90.43	71.73	81.50	91.19
F1-Score (%)	73.60	85.30	95.28	90.41	71.77	81.43	91.11
ROC-AUC (%)	80.30	91.49	98.62	97.01	89.06	91.88	95.49
Precision (%)	75.54	82.81	96.12	90.88	71.85	84.31	91.44
Recall (%)	69.87	72.92	97.95	87.45	71.73	83.04	91.67

Table 2: LightGBM performance metrics (mean of 8-fold cross-validation) across the seven benchmark datasets, as reported in the article.

(b) Critical Evaluation

The reported metrics are broadly realistic and credible rather than inflated: performance is not uniformly high across all datasets — LightGBM's accuracy dips to about 71.7% on Eye Movements and 73.6% on the Cardiovascular Disease dataset, showing the authors did not selectively report only their strongest results. The use of standard deviations alongside every mean score, and the repeated 8-fold cross-validation protocol, also increases confidence that the numbers are not artefacts of a single favourable split. Potential biases remain: the Arcene and Prostate datasets have only 200 and 102 samples respectively against thousands of features, meaning cross-validation folds are small and high-variance, so the near-90%+ scores there should be treated cautiously despite the reported error bars. There is also a possible tuning-effort bias, as noted above, since GBDT and traditional models are cheaper to optimise exhaustively than deep networks under a fixed budget. Additional experiments that would strengthen the findings include an external (out-of-institution) validation set for the clinical datasets, statistical significance testing (e.g., paired t-tests) between the top three GBDT ranks, and an ablation isolating the effect of hyperparameter search budget across model families.

(c) Real-World Applicability

GBDT-based classifiers are well suited to deployment in clinical decision-support tools — for example, flagging early-stage cardiovascular risk or supporting Parkinson's screening from voice recordings — because they combine strong accuracy with low computational cost and reasonable interpretability via feature importance. Scaling to industry settings faces several challenges: data availability is a persistent bottleneck since high-quality labelled medical records are costly and privacy-restricted to obtain; class imbalance and small sample sizes (as in Parkinson's or Prostate) can undermine generalisation once models leave the benchmark setting; and ethical/regulatory issues arise around patient data privacy, model explainability requirements for clinical use, and the risk of embedding historical biases (e.g., demographic underrepresentation) into diagnostic predictions.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights

- Insight 1 — Ensembles vs. single trees: A single Decision Tree ranked worst overall (12.714) while GBDT ensembles ranked best, directly illustrating the CO4 concept that a single inductive tree is prone to high variance/overfitting, and that ensemble techniques (boosting, bagging) systematically reduce this weakness — reinforcing the course's coverage of Decision Trees and ensemble-based Inductive Learning.
- Insight 2 — Evaluation protocol matters as much as the algorithm: The article's use of stratified k-fold cross-validation and ROC-AUC (rather than raw accuracy alone) for imbalanced medical data connects to the CO4 topic of building and validating classification/decision-making models responsibly, showing that metric choice can change which model appears 'best.'

- Insight 3 — No single model dominates universally: Deep learning models like TabNet occasionally beat GBDT on individual datasets (e.g., Parkinson's), which reinforces the Statistical Learning Methods theme that model selection is dataset-dependent, and that comparative, empirical evaluation — not theoretical superiority alone — should guide the choice of a classifier in practice.

(b) Proposed Extension

Proposed Experiment: Extend the study by applying the same GBDT vs. deep-learning vs. traditional-ML benchmarking protocol to a federated learning setting across multiple hospitals, where each institution's data stays local and only model updates are shared, using a new, larger multi-institution disease dataset (e.g., diabetic retinopathy screening records) rather than the largely single-source datasets used in the original study.

Feasibility: This is feasible because federated implementations of GBDT (e.g., federated XGBoost) and federated deep learning frameworks already exist and integrate with common ML libraries, and the same 8-fold cross-validation and ROC-AUC evaluation protocol from the original paper could be reused with minimal modification, only adding a per-institution aggregation step.

Expected Impact: Such an experiment would test whether GBDT's advantage over deep learning persists once data is distributed and privacy-preserving constraints are added, which is the realistic deployment scenario for clinical AI; it would also directly address the real-world data-availability and privacy challenges identified in Task 3(c), producing findings with stronger practical relevance for hospital adoption than a single-institution benchmark.

Rubric Self-Check

Understanding of Article (20%): The review demonstrates in-depth understanding of the article's purpose (comparing GBDT, traditional ML, and tabular deep learning for medical diagnosis), its core arguments, and its methodology.

Summary (20%): Task 1 provides a concise, accurate 150–200 word summary covering the objective and identified research gap.

Critical Analysis (25%): Task 3(b) and 2(b) provide reasoned critique of potential biases (small-sample datasets, tuning-effort asymmetry) rather than simply restating the results.

Use of Evidence (15%): Tables 1 and 2 reproduce the article's actual reported average ranks and LightGBM metrics across all seven datasets, with in-text reference to specific figures.

Organization (20%): The response follows the exact Task 1–4 structure requested, with clear headings and logical flow from summary to methodology to results to reflection.